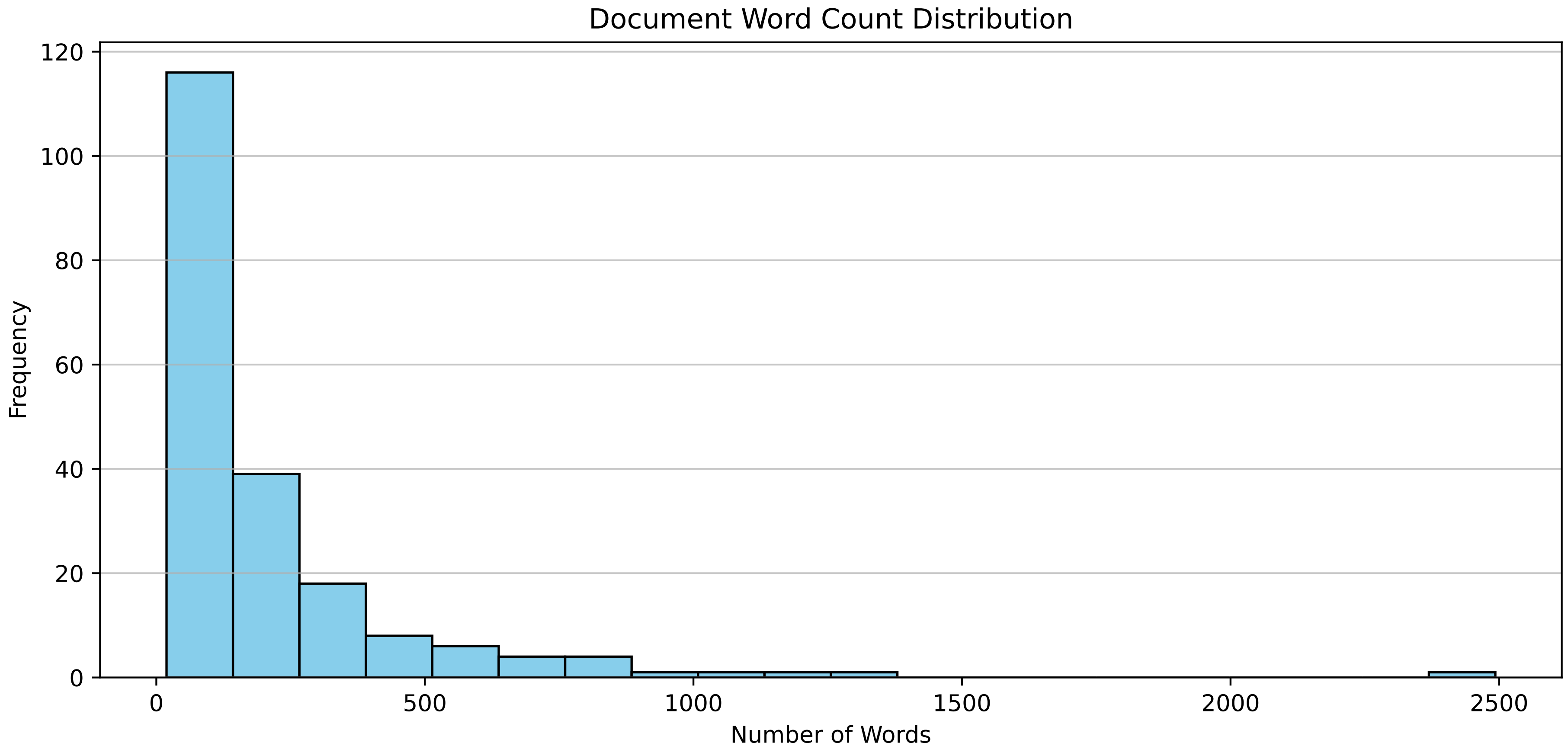
Data

Corpus Summary Statistics:
• Total Documents: 200
• Average Word Count: 207.8 words
• Min Word Count: 19 words
• Max Word Count: 2493 words

Preprocessing Steps:
• Lowercasing & punctuation removal
• English stop-word filtering
• Frequency bounds: min_df=2, max_df=0.95



Methodology & Results

METHODOLOGY:

1. Representation: TF-IDF (Term Frequency-Inverse Document Frequency) was used to vectorize the text, naturally down-weighting ubiquitous vocabulary.

2. Dimension Reduction: Truncated SVD (Latent Semantic Analysis) reduced the highly sparse text vectors down to 50 principal components to isolate latent topics and discard noise.

3. Clustering: K-Means clustering was applied with n_init=20 random restarts to avoid local minima.

4. Libraries Used: pandas, numpy, scikit-learn, matplotlib.

ASSUMPTIONS:

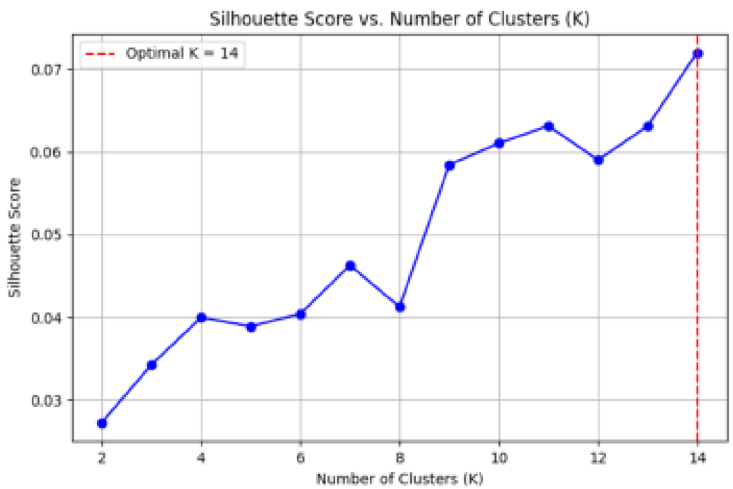
- Documents strictly belong to exactly one topic (Hard Clustering assumption).
- TF-IDF assumes bag-of-words (word order does not dictate topic).

RESULTS:

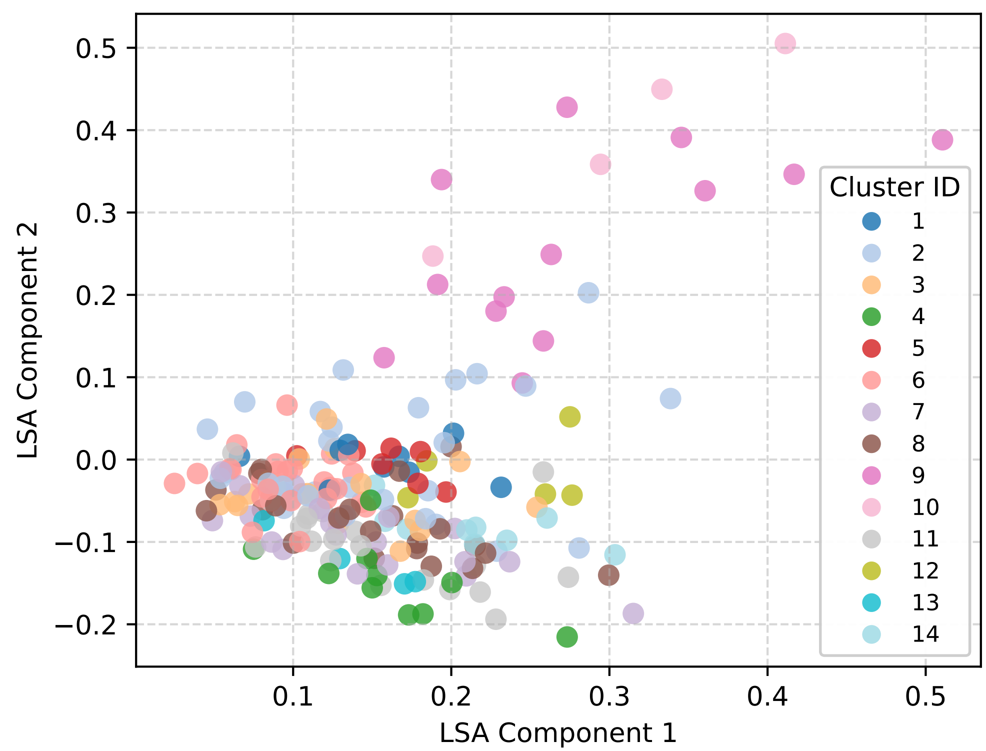
- Optimal Number of Topics (K-hat) = 14
- Chosen by maximizing the average Silhouette Score over a search grid of K=[2, 15].

Slide 3: Clustering Visualizations & Topics

Optimal K Selection (Silhouette)



2D LSA Projection



Cluster Interpretations (Top TF-IDF Terms):

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Cluster  1: jews, killing, suffered, hamas
Cluster  2: just, don, body, subject
Cluster  3: got, sky, left, rights
Cluster  4: year, won, league, standard
Cluster  5: israel, arab, arabs, israeli
Cluster  6: april, pm, philadelphia, try
Cluster  7: car, cars, price, list
Cluster  8: shuttle, nasa, space, need
Cluster  9: god, jesus, faith, religion
Cluster 10: law, sabbath, ceremonial, paul
Cluster 11: baseball, dealer, hit, season
Cluster 12: turkish, turks, turkey, greeks
Cluster 13: small, growth, rate, rocket
Cluster 14: armenians, armenian, nazi, people
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